Giri's Tech Hub, Pune.

Core Java Machine Test

Batch: July-25

Date: 23/03/2026
Time: 02 to 05.00 Pm.

Instructions:

1. Solve any 9 questions.
2. Input should be from user.
3. Indentation and comments mandatory.
4. Without using any inbuilt functions.

Q1. Write a java program to print this pattern.

```
                A
            #   #   #
        A   B   C   B   A
    #   #   #   #   #   #   #
A   B   C   D   E   D   C   B   A
    #   #   #   #   #   #   #
        A   B   C   B   A
            #   #   #
                A
```

Q2. Write a program to reverse an array using recursion.

Input:
Array:
10 20 30 40 50
Output:
Reversed Array:
50 40 30 20 10

Q3. Write a Java program to find the kth largest element in an array without sorting.

Input:
Enter size: 6
Enter elements: 10 45 23 67 34 67
Enter k value: 2
Output: Second largest element is 45

Q4. Write a program in java to rotate an array by N positions without using other array.

Input:
The given array is: 0 3 6 9 12 14 18 20 22 25 27

From 4th position the values of the array are: 12 14 18 20 22 25 27
Before 4th position the values of the array are: 0 3 6 9
After rotating from 4th position the array is: 12 14 18 20 22 25 27 0 3 6 9

Q5. Write a java program to implement binary search from array.

Input:
Array: 10 20 30 40 50 60
Search: 40
Output: Element found at index 3

Q6. Write a Java program to find the second non-repeating character.

Description: Find the second character appearing once.
Input: Enter string: swiss
Output: Second non repeating character: i

Q7. Write a program to enter the String and sum of the all Digits in a given String
input : abcd123pqe12mn12abc
output is Sum is 147
(note sum of number is 123+12+12 =147)

Q8. Check whether expression has balanced parentheses.
Input:
{[()]}
Output:
Balanced

Q9. Write a Java program to find all anagram pairs.
Description: Find anagram pairs from list.
Input: Enter words: cat tac act dog god
Output:
cat tac
cat act
tac act
dog god

Q10.Write program to group words by their length using HashMap.
Input:
cat dog elephant tiger lion
Output:
3 → cat dog
4 → lion
5 → tiger
8 → elephant

-----ALL THE BEST-----